

# BUSINESS REQUIREMENTS DOCUMNT

## CUSTOMER TICKET MANAGEMENT SYSTEM



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# 1. Executive Summary

The Customer Ticket Management System is a centralized platform designed to streamline customer issue tracking, ticket assignment, escalation management, and resolution processes. The system aims to improve customer satisfaction, reduce response times, and provide visibility into support operations.

## 2. Business Objectives

1. **Primary Goal:** Develop a centralized platform for efficient ticket creation, tracking, and resolution.
2. **Operational Efficiency:** Automate ticket assignment, prioritization, and escalation to reduce manual effort and improve response times.
3. **Customer Satisfaction:** Enhance customer experience through faster resolutions, transparent ticket tracking, and timely updates.
4. **Performance Monitoring:** Enable managers to track agent productivity, ticket resolution metrics, and overall support performance.

## 3. As-Is and To-Be Analysis

### 1. As-Is

#### 3.1 As-Is (Current State)

- **Customer complaints arrive through email, calls, and WhatsApp:** Customer issues are received through multiple communication channels, making it difficult to manage and track requests efficiently.
- **Tickets are tracked manually in Excel:** Support teams maintain ticket records in spreadsheets, which increases manual effort and the possibility of errors.
- **No centralized tracking:** There is no single platform where all customer issues can be monitored and managed in one place.
- **Delays in ticket assignment:** Tickets are assigned manually, resulting in delays and uneven workload distribution among support agents.
- **Difficult to monitor agent performance:** Managers have limited visibility into ticket handling, response times, and agent productivity.

## Problems

- **Lost tickets:** Customer requests may be overlooked or missed due to manual tracking processes.
- **Duplicate tickets:** The same issue may be recorded multiple times, causing confusion and inefficiency.
- **Delayed responses:** Manual processes increase response and resolution times, impacting customer satisfaction.
- **Lack of reporting:** The organization lacks meaningful reports and insights to monitor support performance and trends.

## 3.2 To-Be (Future State)

- **Automatically create tickets:** The system will generate unique ticket IDs and store all customer requests in a centralized database.
- **Assign tickets to agents:** Tickets will be automatically assigned based on predefined rules, workload, or expertise.
- **Track ticket status:** Support teams and customers can monitor ticket progress from creation to closure.
- **Escalate unresolved tickets:** Tickets exceeding SLA timelines will be escalated to higher support levels automatically.
- **Generate reports and dashboards:** Management will have access to real-time reports and dashboards for performance monitoring.

## 3.3 User Experience Improvements

- **Customers can raise tickets easily:** Customers can submit issues through a simple and structured ticket creation process.
- **Customers can track ticket status:** Customers will have visibility into the progress and current status of their requests.
- **Customers receive notifications:** Automated updates will keep customers informed about ticket progress and resolution.
- **Agents can view assigned tickets:** Support agents can access and manage all tickets assigned to them from a single dashboard.
- **Agents can update progress:** Agents can update ticket status, add comments, and communicate resolution updates.
- **Agents can close resolved tickets:** Once an issue is resolved, agents can close the ticket and record resolution details.

3.4 Operational Efficiency

- **Automated workflows:** The system will automate ticket creation, assignment, escalation, and notifications, reducing manual intervention.
- **Reduced manual work:** Support agents will spend less time on administrative tasks and more time resolving customer issues.
- **Better resource allocation:** Tickets can be distributed based on agent workload, skill set, or availability to ensure balanced utilization.
- **Faster resolutions:** Automated processes and centralized tracking will help reduce ticket resolution times and improve service quality.

3.5 Future Scalability

- **AI chatbot integration:** Future versions may include AI-powered chatbots to handle common customer queries and reduce support workload.
- **WhatsApp integration:** Customers will be able to raise and track tickets directly through WhatsApp for greater convenience.
- **Knowledge Base:** A self-service knowledge base can be introduced to help customers find solutions without contacting support.
- **SLA automation:** The system can automatically monitor SLA timelines and trigger alerts or escalations when deadlines are at risk.

Key Differences Between As-Is and To-Be States

| Aspect                 | As-Is                                                                      | To-Be                                                                            |
|------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Ticket Creation        | Customer issues are received through emails, calls, and WhatsApp manually. | Tickets are automatically created and recorded in the system with unique IDs.    |
| Ticket Tracking        | Tickets are tracked manually using Excel sheets.                           | All tickets are tracked through a centralized dashboard.                         |
| Ticket Assignment      | Managers or agents manually assign tickets.                                | Tickets are automatically assigned based on predefined rules and workloads.      |
| Ticket Visibility      | Limited visibility into ticket status and ownership.                       | Real-time visibility of ticket status, ownership, and progress.                  |
| Customer Communication | Customers need to follow up manually for updates.                          | Automated notifications keep customers informed throughout the ticket lifecycle. |

| Aspect                        | As-Is                                              | To-Be                                                                |
|-------------------------------|----------------------------------------------------|----------------------------------------------------------------------|
| Agent Productivity Monitoring | Difficult to measure agent performance accurately. | Managers can track agent performance through dashboards and reports. |
| Scalability                   | Difficult to handle increasing ticket volumes.     | System can support growing customer and ticket volumes efficiently.  |

## 4. Scope

The scope of the **Customer Ticket Management System (CTMS)** outlines the key features and functionalities to be delivered in the initial phase of the project. The system is designed to provide a centralized platform for managing customer support requests, improving ticket resolution processes, enhancing customer satisfaction, and enabling efficient monitoring of support operations. The initial release focuses on core ticket management functionalities while supporting future business growth and system expansion.

### 4.1. In-Scope Features

The following features are within the scope of the CTM System and will be implemented in this phase:

#### Ticket Management

- **Create Ticket:** Customers can raise a new support request by providing issue details. The system will generate a unique ticket ID for tracking purposes.
- **Edit Ticket:** Authorized users can modify ticket details such as description, priority, or category before resolution.
- **Close Ticket:** Support agents can close a ticket once the issue has been resolved and confirmed with the customer.
- **Reopen Ticket:** Customers or agents can reopen a closed ticket if the issue persists or requires further investigation.

#### Assignment

- **Auto-Assignment:** The system will automatically assign tickets to available agents based on predefined rules, workload, or expertise.
  - **Manual Assignment:** Managers or team leads can manually assign tickets to specific agents when required.
-

## Escalation

- **Level 1 Escalation:** Tickets that are not addressed within the defined timeframe will be escalated to the immediate supervisor.
- **Level 2 Escalation:** Unresolved tickets will be further escalated to senior support teams for faster resolution.
- **Level 3 Escalation:** Critical or high-priority issues will be escalated to management or specialized teams for immediate attention.

## Reporting

- **Open Tickets:** The system will provide reports showing all currently active and unresolved tickets.
- **Closed Tickets:** Users can view reports containing completed and resolved support requests.
- **Agent Performance:** Managers can monitor key performance metrics such as resolution time, ticket volume, and agent productivity.

## Notifications

- **Email Alerts:** Automated email notifications will be sent for ticket creation, assignment, escalation, and closure events.
- **Status Updates:** Customers and support agents will receive updates whenever there is a change in ticket status.

## 4.2 Out-of-Scope

- **Payment Processing:** The system will not support payment collection, refunds, or financial transaction management.
- **Live Chat Support:** Real-time chat functionality between customers and support agents is not included in the initial release.
- **AI Chatbot:** AI-powered virtual assistant capabilities will not be implemented in the current project scope.

**Social Media Support:** The system will not integrate with social media platforms such as Facebook, Instagram, or X for ticket creation and management.

# 5. Stakeholders

**1. Business Management:** Responsible for approving project objectives, monitoring performance, and ensuring business goals are achieved.

**2. Customers:** Users who raise support tickets, track issue status, and provide feedback on resolutions.

**3. Support Agents:** Responsible for handling, updating, and resolving customer tickets within SLA timelines.

**4. Support Managers/Admins:** Manage ticket assignments, escalation rules, user access, and performance monitoring.

**5. Development Team:** Responsible for designing, developing, testing, and maintaining the CTM system.

**6. QA/Testers:** Responsible for validating system functionality and ensuring requirements are met before deployment.

## 6. User Roles & Permissions

### Customer

Customers can create support tickets, track ticket progress, and receive updates regarding the status and resolution of their issues.

### Support Agent

Support Agents are responsible for handling assigned tickets, updating ticket statuses, communicating with customers, and resolving issues within defined SLA timelines.

### Manager

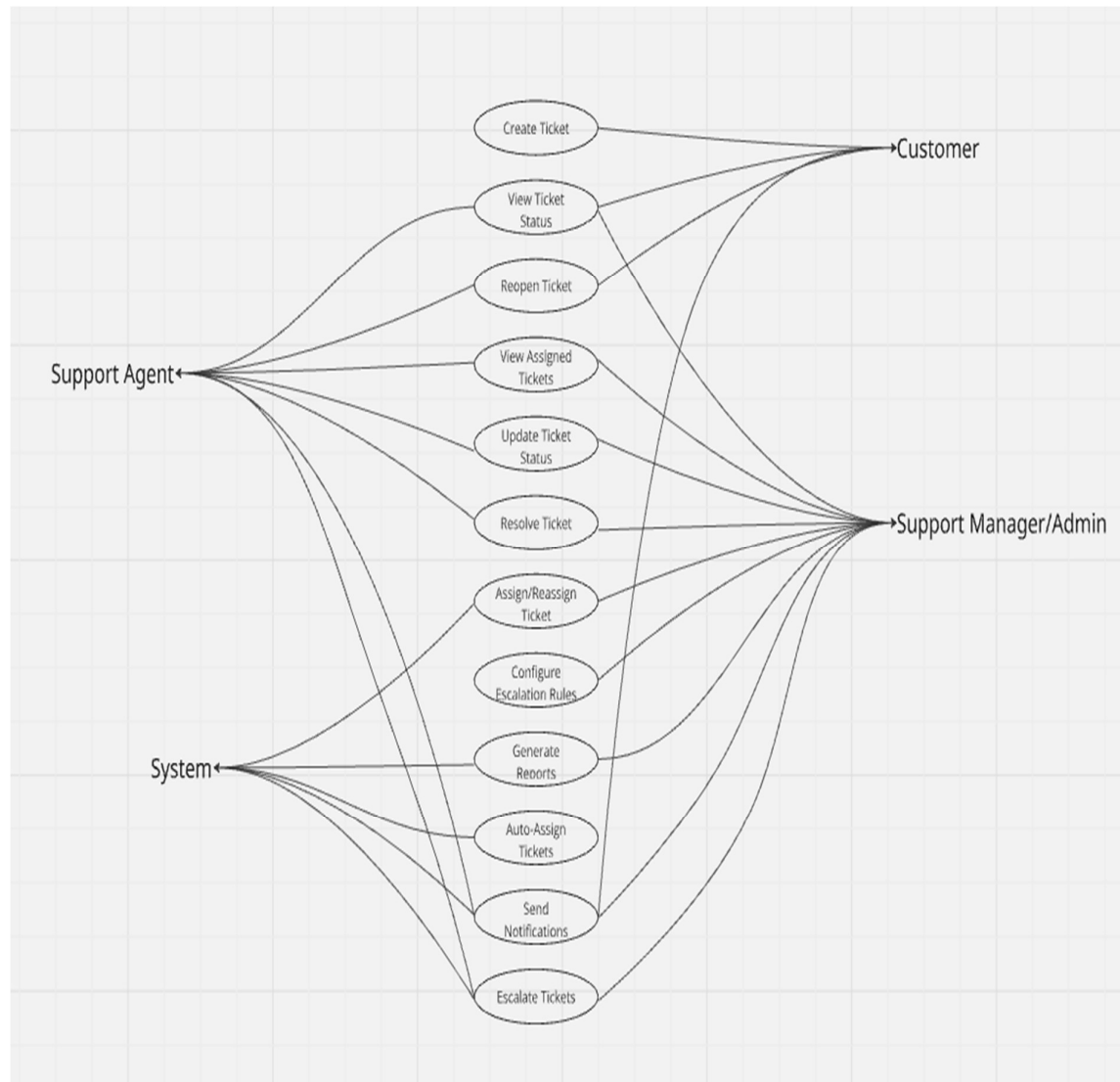
Managers oversee support operations by assigning tickets, monitoring team performance, reviewing escalations, and generating reports for operational analysis.

### Admin

Administrators manage user accounts, configure system settings, maintain ticket categories, and ensure the smooth functioning of the Customer Ticket Management System.



## 7. Use Case Diagram:



## 8. Functional Requirements

Functional requirements describe the specific functions and actions that the **Customer Ticket Management System** must perform to support ticket creation, assignment, tracking, escalation, resolution, reporting, and notification processes. These requirements ensure seamless collaboration between customers, support agents, and administrators while improving service efficiency and customer satisfaction.

**1. Ticket Management:** The system shall provide functionality for customers to create, track, and manage support tickets throughout their lifecycle.

- **FR1.1:** Customers must be able to create a support ticket by providing details such as issue category, subject, description, and attachments.
- **FR1.2:** The system shall generate a unique Ticket ID for every ticket created.
- **FR1.3:** Customers must be able to view the current status of their submitted tickets.
- **FR1.4:** Authorized users shall be able to edit ticket details before resolution.
- **FR1.5:** Customers or support agents shall be able to reopen a ticket if the issue remains unresolved.

**2. Ticket Assignment:** The system shall support automatic and manual assignment of tickets to support agents.

- **FR2.1:** The system must automatically assign tickets based on agent workload and availability.
- **FR2.2:** Managers shall be able to manually assign or reassign tickets when required.
- **FR2.3:** The system shall maintain assignment history for audit and tracking purposes.

**3. Ticket Status Management:** The system shall track tickets using predefined statuses.

- **FR3.1:** The following ticket statuses shall be supported:

Open

In Progress

Pending

Resolved

Closed

- **FR3.2:** Support agents shall be able to update ticket status throughout the resolution process.
- **FR3.3:** Customers shall receive notifications whenever the ticket status changes.

**4. Escalation Management:** The system shall automatically escalate unresolved tickets.

- **FR4.1:** The system must support multiple escalation levels based on business rules.
- **FR4.2:** Tickets exceeding SLA timelines shall be automatically escalated.
- **FR4.3:** Managers shall receive notifications for escalated tickets.

**5. Reporting and Analytics:** The system shall provide reporting capabilities for monitoring support operations.

- **FR5.1:** Managers must be able to generate Daily Ticket Reports.
- **FR5.2:** Managers must be able to generate Weekly Ticket Reports.
- **FR5.3:** Managers must be able to generate Monthly Ticket Reports.
- **FR5.4:** The system shall provide Agent Performance Reports including ticket volume and resolution time.
- **FR5.5:** Dashboards shall display open, closed, pending, and escalated tickets.

**6. Notification Management:** The system shall notify users regarding important ticket activities.

- **FR6.1:** Email notifications shall be sent upon ticket creation.
- **FR6.2:** Notifications shall be sent when tickets are assigned or reassigned.
- **FR6.3:** Notifications shall be sent when tickets are escalated.
- **FR6.4:** Customers shall receive notification upon ticket resolution and closure.

## 9. Non-Functional Requirements

### 9.1 Performance

- **NFR1.1:** The system should load major pages within 3 seconds under normal operating conditions.
- **NFR1.2:** The system should support simultaneous access by multiple users without performance degradation.

## 9.2 Usability

**NFR2.1:** The user interface should be simple and intuitive for customers, support agents, and managers.

**NFR2.2:** Users should be able to create or update tickets with minimal training.

## 9.3 Security

**NFR3.1:** The system shall implement role-based access control (RBAC).

**NFR3.2:** User data and ticket information shall be protected using encryption.

**NFR3.3:** Only authorized users shall be able to access sensitive ticket information.

## 9.4 Scalability

**NFR4.1:** The system should support more than 10,000 tickets per month without significant performance impact.

**NFR4.2:** The architecture should allow future integration with chatbots, WhatsApp, and CRM systems.

## 9.5 Reliability

**NFR5.1:** The system shall maintain 99.9% uptime during business operations.

**NFR5.2:** Daily automated backups shall be performed to prevent data loss.

**NFR5.3:** The system shall maintain activity logs for audit and recovery purposes.

# 10.Context Diagram

### Customer

The customer interacts with the system to create support tickets, track ticket status, and receive updates regarding issue resolution.

### Email System

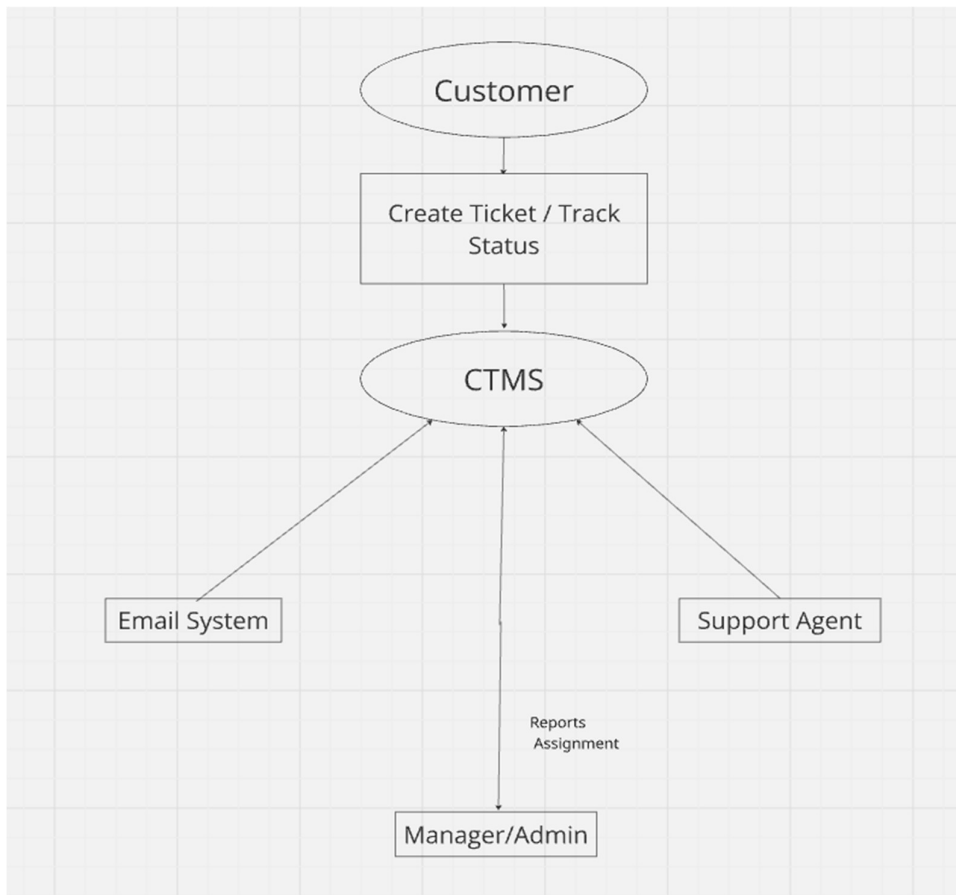
The email system is used to send automated notifications related to ticket creation, assignment, escalation, and closure.

### Support Agent

Support agents access the system to manage assigned tickets, update ticket status, and resolve customer issues.

**Management**

Management uses the system to monitor support operations, track agent performance, and generate reports for decision-making.



## 11. Assumptions & Dependencies

**Assumptions****Customers have internet access**

Customers are expected to have a stable internet connection to create, track, and manage support tickets.

**Agents are trained**

Support agents are assumed to have the necessary knowledge and training to use the system effectively.

## **Dependencies**

### **Email Service Integration**

The system depends on email services to send notifications, ticket updates, and escalation alerts.

### **Database Availability**

The system relies on a database to store and retrieve ticket information, user details, and reports.

## **Risks & Constraints**

### **Risks**

#### **User Adoption Issues**

Users may be resistant to adopting the new system, which could affect its successful implementation.

#### **System Downtime**

Unexpected outages may impact ticket management and customer support operations.

#### **Data Migration Challenges**

Transferring existing ticket data from legacy systems may lead to data loss or inconsistencies.

### **Constraints**

#### **Budget Limitations**

Project features and timelines may be influenced by the available budget and resources.

#### **Timeline Restrictions**

The project must be delivered within the agreed schedule, which may limit the scope of enhancements.

## 12. Timeline & Milestones

| Milestone                  | Timeline      |
|----------------------------|---------------|
| Project Kick-off           | [Insert Date] |
| Completion of UI/UX Design | [Insert Date] |
| Development Start          | [Insert Date] |
| Development Completion     | [Insert Date] |
| User Testing & Feedback    | [Insert Date] |
| UAT Sign-off               | [Insert Date] |
| Production Deployment      | [Insert Date] |
| Project Closure            | [Insert Date] |

## 13. Success Metrics (KPIs)

- Ticket Resolution Time
- First Response Time
- Customer Satisfaction Score
- Ticket Backlog
- Agent Productivity

## 14. Appendices

### 1. Glossary of Terms

| Term | Definition                        |
|------|-----------------------------------|
| CTMS | Customer Ticket Management System |

| Term       | Definition                                                        |
|------------|-------------------------------------------------------------------|
| SLA        | Service Level Agreement                                           |
| Ticket     | A support request raised by a customer                            |
| Escalation | Process of forwarding unresolved tickets to higher support levels |
| Admin      | User responsible for managing system configuration and access     |
| UAT        | User Acceptance Testing                                           |
| KPI        | Key Performance Indicator                                         |
| RBAC       | Role-Based Access Control                                         |

**2. Additional Notes**

- Future releases may include AI-powered chatbots for handling common customer queries.
- Advanced analytics and SLA automation features may be added to enhance support operations and performance monitoring.